Solutions, NUP Selection to IMC 2026

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Problem 1. (IMC 2022, Day 1, Problem 1) $f(x) \cdot f(1-x) \equiv 1$ on $[0,1]$, $f > 0$ integrable. Show $\int_0^1 f \geq 1$.
Solution. By AM–GM,

$$f(x) + f(1-x) \geq 2\sqrt{f(x)f(1-x)} = 2.$$

Integrating on $[0, 1/2]$,

$$\int_0^1 f = \int_0^{1/2} f(x)\,dx + \int_0^{1/2} f(1-x)\,dx = \int_0^{1/2} \big(f(x) + f(1-x)\big)dx \geq \int_0^{1/2} 2\,dx = 1. \quad \square$$

Problem 2. (IMC 2020, Day 1, Problem 2) $\mathrm{rk}(AB - BA + I) = 1 \implies \mathrm{trace}(ABAB) - \mathrm{trace}(A^2B^2) = \frac{1}{2}n(n-1)$.
Solution. Set $X = AB - BA$. By cyclicity of trace,

$$\mathrm{trace}(X^2) = \mathrm{trace}(ABAB) - \mathrm{trace}(AB^2A) - \mathrm{trace}(BA^2B) + \mathrm{trace}(BABA) = 2\,\mathrm{trace}(ABAB) - 2\,\mathrm{trace}(A^2B^2),$$

so it suffices to show $\mathrm{trace}(X^2) = n(n-1)$.

The hypothesis says $X + I$ has rank 1, hence eigenvalue 0 with multiplicity $n-1$. Therefore X has eigenvalue -1 of multiplicity $n-1$, and the remaining eigenvalue equals $\mathrm{trace}(X) - (n-1)\cdot(-1) = 0 + (n-1) = n-1$ (since $\mathrm{trace}(AB - BA) = 0$). Hence

$$\mathrm{trace}(X^2) = (n-1)\cdot(-1)^2 + 1\cdot(n-1)^2 = n(n-1). \quad \square$$

Problem 3. (IMC 2021, Day 1, Problem 3) Find $\sup\{d : d \text{ is good}\}$ where *good* means there is $(a_n) \subset (0,d)$ such that $a_1, \dots, a_n$ partitions $[0,d]$ into $n+1$ pieces of length $\leq 1/n$ each.
Solution. *Answer:* $d^\star = \ln 2$.

Upper bound $d^\star \leq \ln 2$. Fix n and order the lengths $0 \leq \ell_1 \leq \cdots \leq \ell_{n+1}$ of the segments after step n. After placing the next k points $a_{n+1}, \dots, a_{n+k}$, at least $n+1-k$ of the original segments are still untouched, so $\ell_{n+1-k} \leq 1/(n+k)$. Therefore

$$d = \sum_{i=1}^{n+1} \ell_i \leq \frac{1}{n} + \frac{1}{n+1} + \cdots + \frac{1}{2n} \xrightarrow[n\to\infty]{} \ln 2.$$

Lower bound $d^\star \geq \ln 2$. The identity

$$\ln 2 = \sum_{i=1}^{n} \ln\left(1 + \frac{1}{n+i-1}\right)$$

gives a partition of $[0, \ln 2]$ into n parts, each of length $\ln(1 + 1/n) < 1/n$. Going from n to $n+1$ parts replaces $\ln(1+1/n)$ by $\ln(1 + 1/(2n)) + \ln(1 + 1/(2n+1))$, which equals $\ln(1 + 1/n)$; this corresponds to inserting one new point inside the segment of length $\ln(1+1/n)$. Iterating produces an admissible sequence on $[0, \ln 2]$. The first terms are $a_1 = \ln\frac{3}{2}$, $a_2 = \ln\frac{5}{4}$, $a_3 = \ln\frac{7}{4}$, $a_4 = \ln\frac{9}{8}$, ... $\square$

Problem 4. (IMC 2020, Day 1, Problem 4) Real polynomial with $p(x+1) - p(x) = x^{100}$. Show $p(1-t) \geq p(t)$ for $0 \leq t \leq 1/2$.
Solution. Since $p(x+1) - p(x) = x^{100}$, comparing leading terms forces $\deg p = 101$ and the leading coefficient $= 1/101$; the coefficient a_0 is free, but does not affect $p(1-t) - p(t)$. So choose the unique solution p_n of $p_n(x+1) - p_n(x) = x^n$ with $p_n(0) = p_n(1) = 0$. Differentiating gives the recursion $p_n'(x) = np_{n-1}(x) + c_{n-1}$ for some constant.

Claim: for every $n \geq 1$ and all real x,

$$p_{2n+1}(x) = p_{2n+1}(1-x), \qquad p_{2n}(x) + p_{2n}(1-x) = 0, \qquad c_{2n} = 0.$$

Proof by induction in n. Base $p_1(x) = \frac{1}{2}x^2 - \frac{1}{2}x$ satisfies $p_1(x) - p_1(1-x) = 0$. For the step, the derivative of $p_{2n}(x) + p_{2n}(1-x)$ equals $2n\,p_{2n-1}(x) + c_{2n-1} - (2n)\,p_{2n-1}(1-x) - c_{2n-1} = 0$, so this is a constant which equals $p_{2n}(0) + p_{2n}(1) = 0$. Then $(p_{2n+1}(x) - p_{2n+1}(1-x))' = (2n+1)(p_{2n}(x) + p_{2n}(1-x)) + 2c_{2n} = 2c_{2n}$, so $p_{2n+1}(x) - p_{2n+1}(1-x) = 2c_{2n}x + b$; evaluating at $0, 1$ gives $b = 0$ and $c_{2n} = 0$.

In particular $p_2''(x) = 2p_1'(x) = 2x-1 < 0$ for $x < 1/2$, so p_2 is strictly concave on $[0, 1/2]$ with $p_2(0) = p_2(1/2) = 0$, hence $p_2(x) > 0$ on $(0, 1/2)$. From $p_4''(x) = 12p_2(x) > 0$ on $(0, 1/2)$, p_4 is strictly convex with $p_4(0) = p_4(1/2) = 0$, so $p_4 < 0$ there. By induction $(-1)^{n-1}p_{2n} > 0$ on $(0, 1/2)$. Therefore p_{100} (with $n = 50$ even) is negative on $(0, 1/2)$, and

$$p(1-t) - p(t) = p_{100}(1-t) - p_{100}(t) = -2\,p_{100}(t) > 0$$

for $0 < t < 1/2$, with equality at $t = 0, 1/2$. $\square$

Problem 5. (IMC 2022, Day 2, Problem 8) Random circle: n blue, k red points; $m := \#$vertices of $F = \mathrm{conv}(\mathrm{red}) \cap \mathrm{conv}(\mathrm{blue})$. Find $[m]$.

Solution. *Answer:*

$$[m] = \frac{2kn}{n+k-1} - \frac{2 \cdot k! \cdot n!}{(k+n-1)!}.$$

Scan the $n+k$ points counterclockwise from a fixed point and read off the colour sequence $(C_1, \dots, C_{n+k})$; regard indices cyclically. Two cases:

(i) Contiguous case: all reds are consecutive (and so are all blues). Then the convex hulls do not intersect, so $m = 0$, but the cyclic colour sequence has exactly 2 colour changes.

(ii) Otherwise (alternating at least twice): every vertex of F is the intersection of a red chord (from a red point to its next red neighbour) with a blue chord; one checks that vertices of F correspond bijectively to colour changes C_iC_{i+1}. So in this case $m = \#(\text{colour changes})$.

In either case

$$m = \#(\text{colour changes}) - 2 \cdot \mathbf{1}[\text{case (i)}].$$

Each cyclic placement of n blues among $n+k$ positions is equally likely, so

$$\Pr[C_i \neq C_{i+1}] = \frac{2nk}{(n+k)(n+k-1)},$$

and there are $n+k$ such pairs cyclically, giving $[\#\text{changes}] = \dfrac{2nk}{n+k-1}$.

The number of cyclic placements giving case (i) equals $n+k$ (choose where the red block starts), out of $\binom{n+k}{n}$ total. So

$$\Pr[\text{case (i)}] = \frac{n+k}{\binom{n+k}{n}} = \frac{n!\,k!}{(n+k-1)!}.$$

Combining,

$$[m] = \frac{2nk}{n+k-1} - 2 \cdot \frac{n!\,k!}{(n+k-1)!}. \quad \square$$